INCH-POUND

MIL-C-19759E
12 December 1989
SUPERSEDING
MIL-C-19759D
26 October 1977

MILITARY SPECIFICATION

CLOTH, COATED, TWILL, NYLON (LOW COUNT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for chloroprene coated nylon cloths.

1.2 Classification. The coated cloths shall be of the following types as specified (see 6.2):

Type I - Nylon cloth coated on one side
Type II - Nylon cloth coated on both sides

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

MILITARY

MIL-C-19256 - Cloth, Twill, Nylon (Low Count, 3.5 ounce)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-595 - Color

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1487 - Glossary of Cloth Coating Imperfections

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-9099.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D1424 Test for Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

TECHNICAL MANUAL OF THE AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS

Method No. 8 Colorfastness to Crocking: AATCC Crockmeter Method

(Application for copies should be addressed to the AATCC National Headquarters, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Materials.

3.3.1 Base cloth. The base cloth shall conform to MIL-C-19256, except that for the type II coated cloth the color shall be natural white.

3.3.2 Coating compound. The coating shall be chloroprene rubber suitably compounded; plasticized; and pigmented in the colors specified in 3.3.3.1 and 3.3.3.2. No elastomer other than chloroprene shall be used (see 4.4.1).

3.3.3 Coated cloth.

3.3.3.1 Type I cloth. The type I cloth shall be coated on one side of the base cloth only with approximately 4.0 ounces per square yard (135.6 g/m^2) of coating compound (see 4.4.1), pigmented black.

3.3.3.2 Type II cloth. The type II cloth shall be coated on one side of the base cloth with approximately 4.0 ounces per square yard (135.6 g/m^2) of coating compound (see 4.4.1), pigmented black. The other side of the base cloth shall be coated with approximately 1.5 ounces per square yard (50.9 g/m^2) of coating compound (see 4.4.1), pigmented green to match shade number 34096 of FED-STD-595.

3.3.3.3 Types I and II cloths. Unless otherwise specified (see 6.5), the applicable cloth shall be vulcanized.

3.3.3.4 Physical requirements. Following vulcanization, the types I and II cloth shall conform to the physical requirements shown in table I when tested as specified in 4.5.

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TABLE I. Physical requirements

Characteristics	Type I Minimum	Type I Maximum	Type II Minimum	Type II Maximum
Weight, oz/yd^2 (g/m^2)	7.0 (237.3)	9.0 (305.1)	8.5(288.2)	10.5(356.0)
Breaking strength, lbs (N)				
Warp	225 (1001)	-	225 (1001)	-
Filling	210 (934)	-	210 (934	-
Tear strength, grams				
Warp	4500	-	3000	-
Filling	3800	-	3000	-
Hydrostatic resistance, lbs/in^2 (PA)				
Initial	100 (689.4 x 10^3)		100 (689.4 x 10^3)	
After 5 launderings	80 (551.5 x 10^3)		80 (551.5 x 10^3)	
After 80 hours weatherometer	80 (551.5 x 10^3)		80 (551.5 x 10^3)	
Flex stiffness, inch-lbs x 10^{-4} (N/m) at 70°F (21°C)				
Warp	-	9.0 (1.02)	-	14.0 (1.58)
Filling	-	7.0 (0.79)	-	11.0 (1.24)
Flex stiffness, inch-lbs x 10^{-4} (N/m) at -40°F (-40°C)				
Warp	-	25.0 (2.83)	-	35.0 (3.96)
Filling	-	25.0 (2.83)	-	35.0 (3.96)
Adhesion of coating, lbs (N/m) per 2 inch (5.1cm) width	10 (1751)	-	10 (1751)	-
Resistance to heat at 160°F (71°C)	No evidence of tackiness, exudation, or loss of flexibility			
Blocking, scale rating	-	No. 2	-	No. 2

3.3.4 Dusting powder. The types I and II cloth (coated side or sides as applicable), shall be dusted with a suitable dusting medium that does not support mildew growth (see 4.4.1). Only the minimum amount of dusting powder necessary to prevent blocking should be applied. The application of dusting powder shall be controlled during processing or reduced by an after processing brush or wash, so that the coated side or sides of the cloth shall show not less than "good" resistance to dry crocking when tested as specified in 4.5.

3.4 Width. The minimum overall width after selvage trimming shall be as specified (see 6.2). Selvages shall be trimmed to give a straight and uniform edge.

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3.5 Length and put-up. Unless otherwise specified, the finished cloth as applicable shall be put-up in rolls as specified in PPP-P-1136 and furnished in a continuous length. Each length shall be not less than 50 yards (45.7 m) nor more than 150 yards (136.5m), (see 6.2).

3.6 Workmanship. The finished cloth shall conform to the quality established by this document. The occurrence of defects shall not exceed the applicable quality level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. Cloth must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be visually examined for appearance, color, finish, and visual defects (see 4.4.2.1) and shall be tested for physical and chemical properties in accordance with methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. A certificate of compliance shall be submitted for the coating compound specified in 3.3.2, the coating weights for types I and II cloth as applicable in 3.3.3.1 and 3.3.3.2, and for the dusting medium requirements specified in 3.3.4.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2.

4.4.2.1 Yard-by-yard examination. The required yardage of each roll shall be visually inspected on one side only; however, the side examined shall be alternated on every other roll for defects listed below. The same yardage shall be given a through-lighting inspection for pinholes only in accordance with MIL-STD-1487. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth in which case only the more serious defect shall be counted. The sample unit shall be one linear yard (0.9m). The lot size shall be expressed in units of one linear yard (0.9m). The sample size shall be in accordance with inspection level III. The acceptable quality levels (AQL), shall be 1.0 major defects and 4.0 total defects (major and minor combined) per hundred units (yards - meters). The number of rolls selected shall be in accordance with Table II. An approximate equal number of yards (meters) shall be examined from each roll in the sample.

Defects	Classification Major	Minor
Uncoated side (type I cloth)		
Any hole, cut or tear	X	
Smash, i.e., damaged area where the texture has been ruptured	X	
Baggy, ridgy, or wavy	X	
Any abrasion mark or bruise, i.e., area damaged by friction	X	
Any evidence of strike through of coating	X	
Any spot or stain (longest dimension in any direction)		
- up to and including 1/4 inch (6.4 mm) in length		X
- more than 1/4 inch (6.4 mm) in length	X	
Any evidence of a tight pick or end, resulting in a ridge or pucker	X	
Any evidence of woven in waste, e.g., specks or fly neps in colors other than that of the cloth	X	
Any broken or missing yarn:		
- up to and including 1/4 inch (6.4 mm) in length		X
- more than 1/4 inch (6.4 mm) in length	X	
Any area of general uncleanness		X
Knot, larger than C on the Sears Fabric Defect Scale (see 6.6)	X	
Slub, larger than D or 3-1/2 on the Sears Fabric Defect Scale (see 6.6)	X	
Off shade, shaded side to side, side to center, end to end, mottled, cloudy, or streaked	X	

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Defects	Classification Major	Minor
Coated side or sides (types I and II cloth)		
Any uncoated area	X	
Any cut, deep scratch, heavily abraded area, hole or tear	X	
Any area where coating is noticeably thinner	X	
Any blister, tunnel, or delamination of coating	X	
Any lump or heavily coated area		X
Crease or wrinkle resulting in doubling or adhesion of surfaces that cannot be corrected by manual pressure	X	
Coating cracked or checked	X	
Coating not applied over full width of base cloth	X	
Any pinhole	X	
Any transparent area or window resulting from improper distribution of pigment	X	
Any evidence of puckiness, waviness, or distortion	X	
Offshade, shaded side to side, side to center, end to end, mottled, cloudy, or streaked	X	
Any area of general uncleanness		X
Any tackiness, coating will adhere and not readily roll	X	
Any objectionable odor other than that which is characteristic of the coating compound		X
Any spot, stain, streak, or imbedded foreign matter:		
- less than 1 inch (2.5 cm) in its longest direction		X
- 1 inch (2.5 cm) or more in its longest direction	X	
Selvages not trimmed off; edges folded, rolled, or not cut straight		X
Any brittle coating (cracks when flexed)	X	

Table II - Sample size

Lot size (yards)	Sample size (rolls)
Up to 1200 inclusive 1/	3
1201 up to and including 3200	5
3201 up to and including 10,000	8
10,001 up to and including 35,000	13
35,001 up to and including 150,000	20
150,001 and over	32

1/ If lot contains fewer than 3 rolls, each roll in the lot shall be examined.

4.4.2.2 Examination for length.

4.4.2.2.1 Individual rolls. During the yard by yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

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4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of the lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

4.4.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of PPP-P-1136 to determine that preservation, packaging, packing, and marking comply with section 5 requirements.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table IV shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final results. The sample unit for test purposes shall be 3 continuous yards (3.2 m) full width of cloth in the vulcanized state. The lot size shall be expressed in units of one yard (0.9m). The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be as indicated in Table III.

Table III - Sample size

Lot size (yards) (meters)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV - Test methods

Characteristic	Requirement Paragraph	Test Method
Weight	3.3.3.4	5041
Breaking strength:	3.3.3.4	5100
Tear strength:	3.3.3.4	ASTM-D1424-63 1/
Hydrostatic resistance:		
Initial	3.3.3.4	5512 2/
After 5 launderings	3.3.3.4	5556 3/4/ 5512 2/
After 80 hrs. weatherometer	3.3.3.4	5804 4/5/ 5512 2/
Flex stiffness at 70°F (21°C):	3.3.3.4	5206 6/ 7/
Flex stiffness at -40°F (-40°C):	3.3.3.4	5206 6/ 8/
Adhesion of coating	3.3.3.4	5970
Resistance to heat	3.3.3.4	5850 9/
Blocking	3.3.3.4	5872
Resistance to dry crocking	3.3.4	AATCC TM 8-1985 10/

1/ Standard instrument with one augmenting weight.
2/ The water pressure shall be applied only to the uncoated side of the type I test specimens or the lighter green coated side of the type II test specimens, whichever is applicable.

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3/ After subjection to laundering or weathering, but prior to hydrostatic testing, the test specimens shall be visually examined for any evidence of cracking, blistering, stiffening, or delamination of the coating from the base cloth. Any evidence of these conditions shall be cause for rejection of the lot.
4/ The cotton laundering procedure shall be followed.
5/ For the type I cloth, the test specimens (coated side) shall face the light source. For the type II cloth, the test specimens (black coated side) shall face the light source.
6/ For the type I cloth, ten (10) test specimens shall be tested and averaged, five with the coated side up and five with the coated side down. For the type II cloth, ten (10) test specimens shall be tested and averaged, five with the face (green coated) side up and five with the face (green coated) side down.
7/ Prior to performing this test, the test specimens shall be conditioned for at least 4 hours at this specified temperature.
8/ Prior to performing this test, the test specimens shall be conditioned for 1 hour at this specified temperature.
9/ The test specimens shall be heated in the oven for 120 hours at 160°F (71°C), and then visually examined.
10/ The test shall be conducted using the blue/black crock cloth.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1136.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1136.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type I coated cloth is intended for use in the manufacture of special cold weather clothing items. The type II coated cloth is intended for use in the manufacture of wet weather clothing items.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number and date of this specification.
b. Type required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
d. When a first article is required (see 3.2 and 4.3). The item will be tested and should be a first article sample. The contracting officer should include specific instruction in acquisition documents regarding arrangements for examinations, quantity, testing, and approval of the first article.
e. Width required (see 3.4).
f. Length required if other than specified (see 3.5).
g. Levels of put-up and preservation, and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids (see 3.1).

* 6.4 Recyled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specificatin.

6.5 Vulcanization. It should be noted that various end items fabricated from these coated cloths require that the material be furnished in the unvulcanized state. The end item procurement documents should therefore be reviewed to determine this requirement. Examples of items that would require that the cloth be furnished unvulcanized are: Overalls, Wet-Weather, MIL-O-22776; Raincoat, Man's (Foul Weather), MIL-R-17657; Hat, Sou'wester, MIL-H-17656; and MIL-P-82277, Parka Wet-Weather.

6.6 Sears Fabric Defect Scales. Sears Fabric Defect Scales are available from Sears, Roebuck and company, Department 817, (Attn: BSC 23-29), Sears Tower, Chicago, IL 60684 (see 4.4.2.1).

* 6.7 Subject term (key word) listing.
Cloth, coated
Cloth, nylon
Cloth, twill
Vulcanization
Wet weather clothing

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* 6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-19759E	CLOTH, COATED, TWILL, NYLON (LOW COUNT)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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